GTM965500P

Test Report

54V Model

GlobTek Engineering

Generated: April 16, 2026

Test Summary

Project: GTM965500P | 54V Model | Total Test Items: 21

Status	OK / Resolved	For Review	Regression Req'd	In Progress	Has Issue	Invalid/ Incomplete Test	Unknown
Count	2	8	0	3	0	0	8

54V

Test Item	Status	Link
Input		
No Load Input Power	Marked for review	View →
Input Voltage	OK / Resolved	View →
Main Output		
Turn-On Delay	Marked for review	View →
Load Regulation	Marked for review	View →
Voltage Ripple	OK / Resolved	View →
Transient Response	Marked for review	View →
Startup Overshoot	Marked for review	View →
Hold-Up Time	Marked for review	View →
Remote-Sense (SNS)	In Progress	View →
Output Voltage Fine Adjustment	Marked for review	View →
Protections		
Over-Power Protection (OPP)	Marked for review	View →

Changes Required

2 changes flagged across all tests.

Voltage	Test	Change Description
54V	Over-Power Protection (OPP) · *Vision314* · *2026-04-01*	COMPONENT | OLD VALUE | NEW VALUE -- | -- | -- C26 | 6.8nF | 6.8nF C27 | 3.3nF | 2.2nF
54V	Output Voltage Fine Adjustment · *Vision314* · *2026-04-06*	Ra8, Ra11 = 200kOhm Ra9, Ra12 = 4kOhm Ra10, Ra13 = 10kOhm Ra14 = 18kOhm

Tests Not Yet Run

10 tests have not yet been run.

54V

Test Item	Status	Link
Protections		
Over-Voltage Protection (OVP)	In Progress	View →
Short-Corcuit Protection (SCP)	In Progress	View →
Environmental / Reliability		
Component Thermal Stress	Unknown	View →
Component Electrical Stress	Unknown	View →
Safety		
Output Touch Current	Unknown	View →
Earth Leakage Current	Unknown	View →
EMC		
Conducted Emissions	Unknown	View →
Radiated Emissions	Unknown	View →
Electrostatic Discharge (ESD)	Unknown	View →
Electrical Fast Transient (EFT)	Unknown	View →

Detailed Test Results

54V

54V	No Load Input Power	OPEN	Marked for review

Specification / Procedure

Spec: < 500mW @ 240Vac (main-output disabled, i.e. remote-off low)

Use Yokogawa WT3000 with 20 second acquisition time (update rate)

Results

Vision314 · 2026-04-15 [LATEST]
560mW

Comments (1)

lauxtim · 2026-04-15 [LATEST]
> [RESULT] 560mW

We could:

1. Change the spec to "500mW typ."...a bit of "specsmanship",
2. Change the spec to "600mW max."
3. Analyze where the 500-600mW is coming from...It does seem a bit high for essentially only the aux supply operating.

500mW (max.) *is* a popular spec among the competition, though I kind of doubt customers actually care...

--

Aux supply has pre-load resistors R71 and R72 to improve its cross regulation performance. About 80mW dissipated in R71 (12^2/ 590*3), and 40mW dissipated in R72 (5^2/590). But we cant' really change the values because they affect the cross-regulation performance.

560 - 120 = 440mW

Startup resistors R59 and R60 should only account for maybe 10mW. Bleeder resistors R1 and R2 about the same.

So where's all this remaining power going...hmm..

We saw those big gulps of 4 or 5W on the Yokogawa, right? That's probably it. That might be indicative of LLC or PFC controller trying to stay afloat via SUPHV startup resistors (which is very lossy, but not an issue because it's only supposed to activate once at startup.)

I think we should try to pin-point who is demanding power when we see the big 4-5W (?) gulps. Let's look at it together. We'll take a look at SUPIC. If it does not seem easily solvable, we'll punt on it an just adjust the spec I think.

54V	Turn-On Delay	OPEN	Marked for review

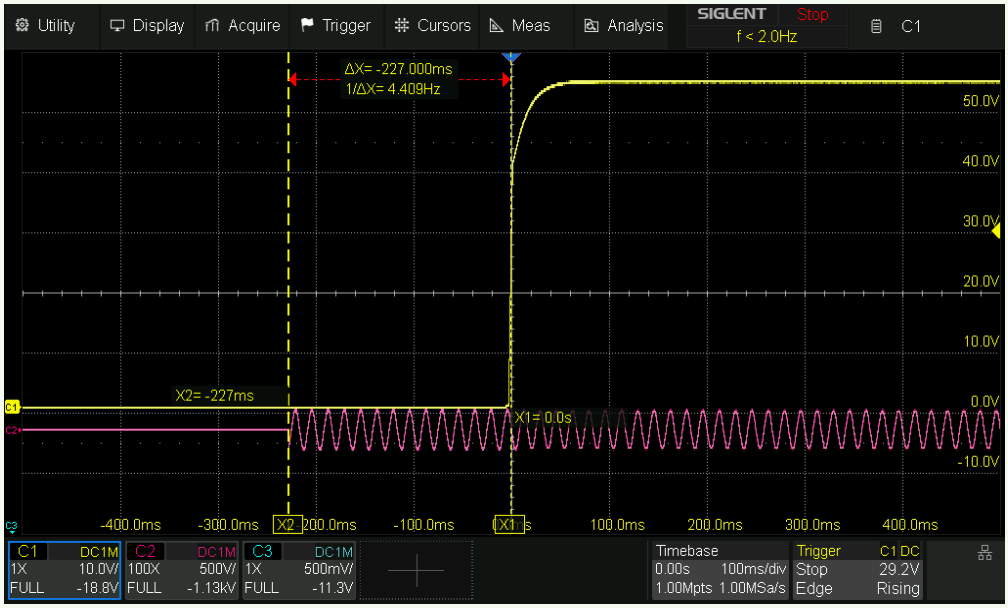
Specification / Procedure

Spec: < 2 seconds @ 115Vac

Measure AC input voltage with differential probe, passive probe on DC output. Check the time between AC turn-on and DC turn-on, at full load.

Results

Vision314 · 2026-04-06 [LATEST]
227ns @ 10.2A



54V	Load Regulation	OPEN	Marked for review

Specification / Procedure

Spec: ± 1.0% (measured at output connector)

Check the stabilized DC output voltage at no-load (0%). Check the stabilized DC output voltage at full load (100%).

% Load regulation = [V(no_load) - V(full_load) / V(full_load)] * 100%

Measure at 90Vac
Measure at 264Vac

Results

Vision314 · 2026-04-06 [LATEST]

Input	V (no load)	V (full load)	Load Regulation (%)
90Vac	54.03	54.03	0.00%
264Vac	54.03	54.03	0.00%

54V	Voltage Ripple	CLOSED	OK / Resolved

Specification / Procedure

Spec: < 1% pk-pk (measured with a 47µF low-ESR cap + 0.1µF ceramic cap, measured @ 20MHz BW)

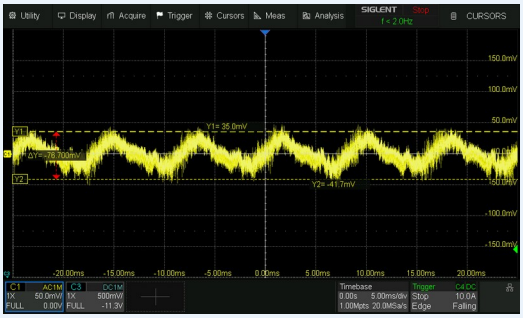
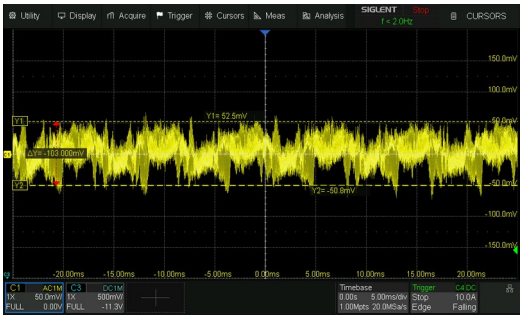
Prior to recording data, check whether 90Vac or 264Vac yields worse results, if at all. Record data using the worst case input voltage.

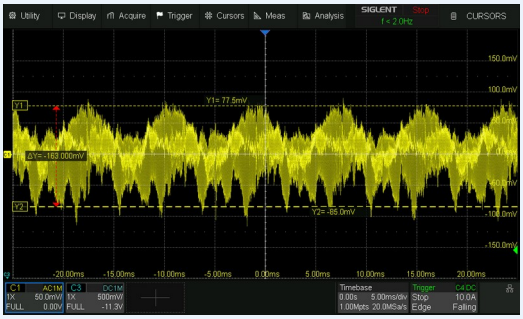
Using the ripple probe fixture and matching 1x probe, connect close to the output connectors on the PCB, measure ripple with oscilloscope. Ripple is inclusive of HF (100KHz) and LF (120Hz) components. Timebase should capture 120Hz component (e.g. 2-5ms/div)

Three conditions, adjust potentiometer accordingly:
(95% * V_out)
(100% * V_out)
(105% * V_out)

Results

Vision314 · 2026-04-07 [LATEST]
1% ripple = 540mV
ALL PASS

Output Voltage	Picture	Measured Ripple
95% Vout		76.7mV
100% Vout		103mV

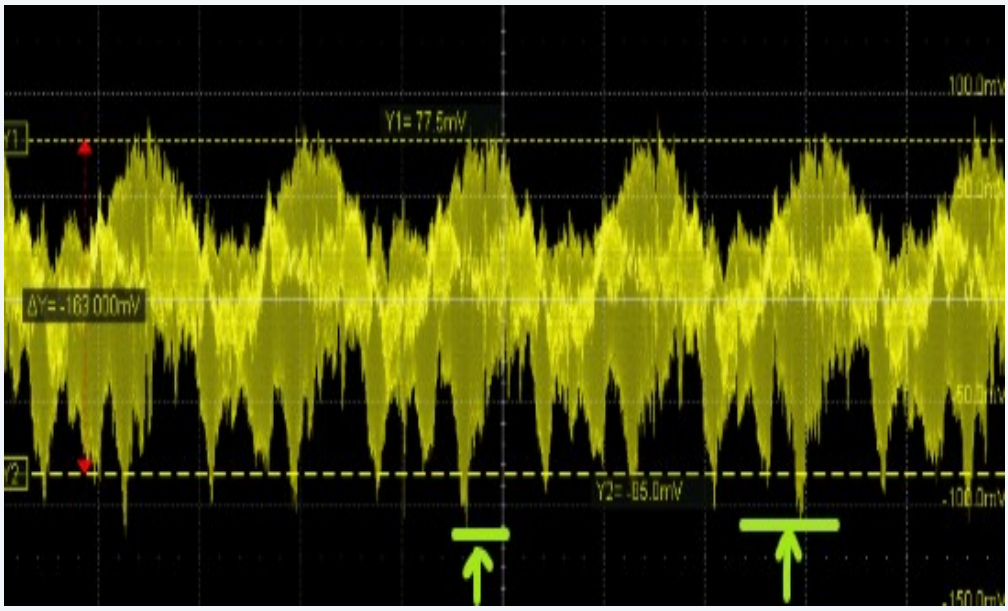
105% Vout		163mV

Comments (1)

lauxtim · 2026-04-14 [LATEST]
@Vision314

Good. For the future, I would include all spurious parts of the waveform as part determining the ripple. I know 95% of the waveform falls within the bounds you show, but it's still a bit subjective.

We are very far away from the limit here so it's not really an issue.



54V	Transient Response	OPEN	Marked for review

Specification / Procedure

Spec: ± 5% deviation max. (with 50% load step), recovery to within ±1% within 1ms

Taken at nominal input voltage (e.g. 115Vac)

Measure with ripple probe fixture PCB, Measure at power supply PCB. Measured with AC coupling.

Step load + load dump:
0 -> 25%
25% -> 50%
50 -> 75%
75 -> 100%

8 measurements total

Results

Vision314 · 2026-04-07 [LATEST]
1% = 540mV
5% = 2.7V

Load Step

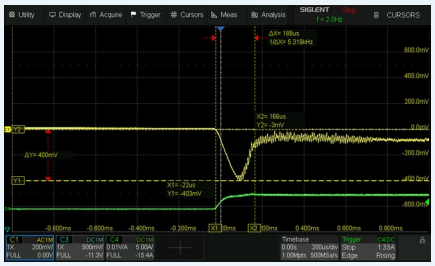
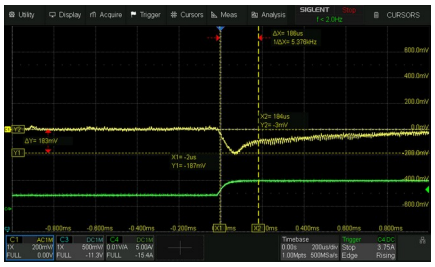
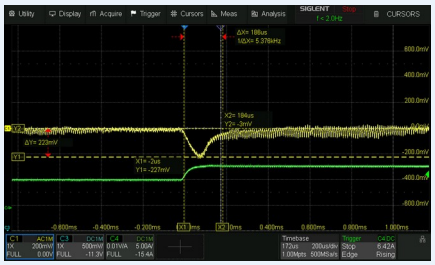
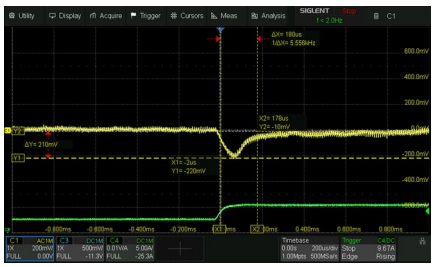
Transition	Deviation	Recovery Time
0A → 2.5A	400mV	<1ms
2.5A → 5.1A	183mV	1ms
5.1A → 7.65A	223mV	<1ms
7.65A → 10.2A	210mV	<1ms

Load Dump

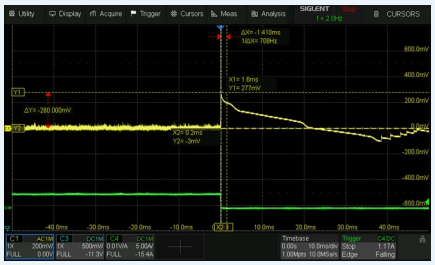
Transition	Deviation	Recovery Time
0A → 2.5A	280mV	1ms
2.5A → 5.1A	200mV	<1ms
5.1A → 7.65A	217mV	<1ms
7.65A → 10.2A	237mV	<1ms

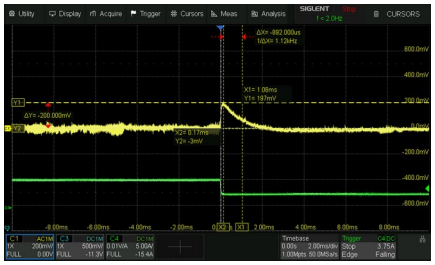
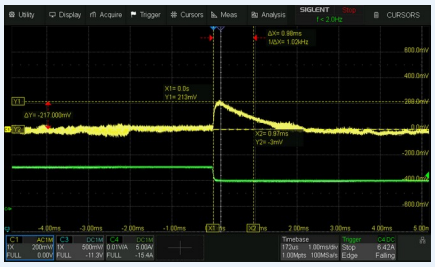
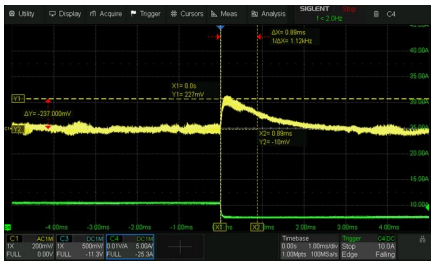
Load Step

Load Step	Picture	Deviation	Recovery Time

0A → 2.5A		400mV	<1ms
2.5A → 5.1A		183mV	1ms
5.1A → 7.65A		223mV	<1ms
7.65A → 10.2A		210mV	<1ms

Load Dump

Load Dump	Picture	Deviation	Recovery Time
0A → 2.5A		280mV	1ms

2.5A → 5.1A		200mV	<1ms
5.1A → 7.65A		217mV	<1ms
7.65A → 10.2A		237mV	<1ms

Comments (1)

lauxtim · 2026-04-14 [LATEST]
See: https://github.com/GlobTek-Engineering/GTM965500P/issues/43#issuecomment-4247254592

54V	Startup Overshoot	OPEN	Marked for review

Specification / Procedure

Spec: <5%

Measure with oscilloscope in DC coupling.

Measure at 90Vac and 264Vac, at:
No load
Full load

4 measurements total

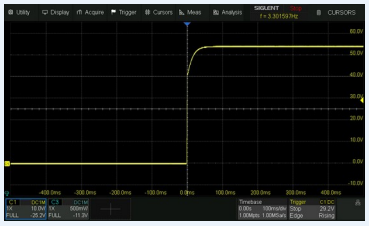
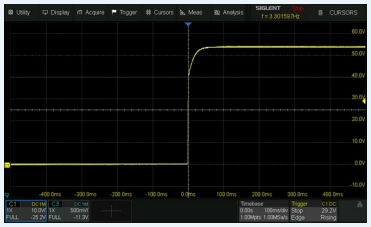
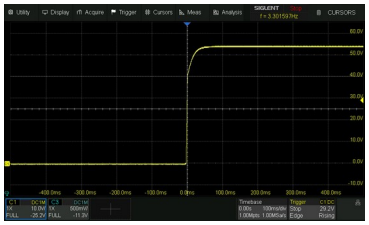
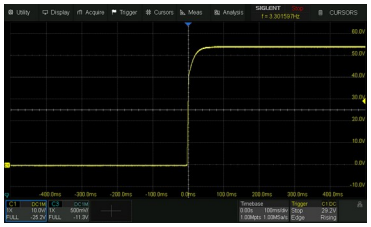
Results

Vision314 · 2026-04-06 [LATEST]

Output Voltage Overshoot

Condition	90 Vac	264 Vac
No Load	0mV	0mV
Full Load	0mV	0mV

Output Voltage Overshoot

Condition	90 Vac	264 Vac
No Load		
Full Load		

54V	Hold-Up Time	OPEN	Marked for review

Specification / Procedure

Spec: 20ms typ. (100% load), 20ms min. (95% load), 50ms typ. (50% load)

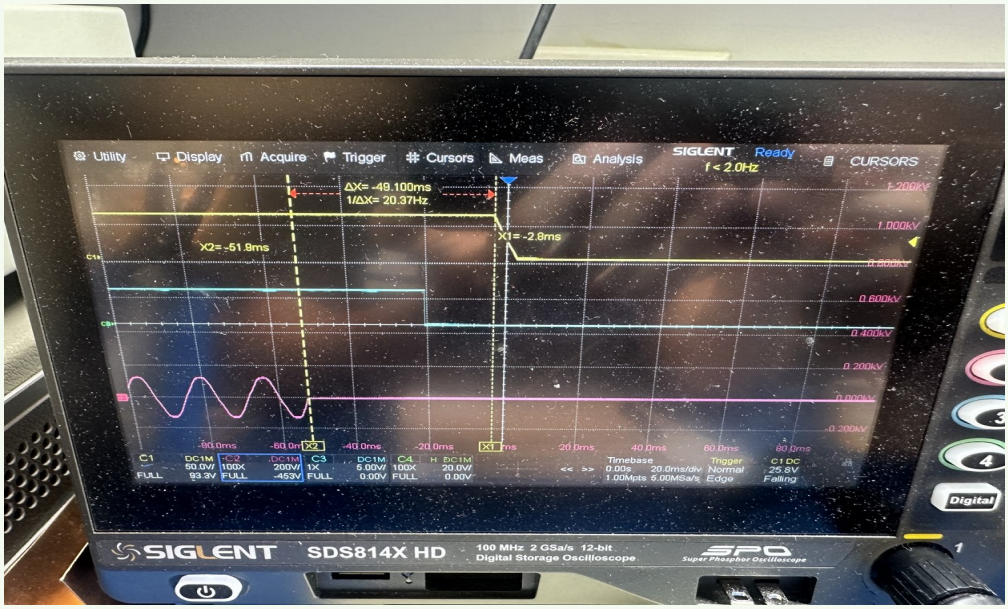
Attach differential probe to AC input. Attach passive probe to output. Set load according to the above. Measure the time between AC turn-off and DC output falling to approx. 95% * nominal V_out.

Results

Vision314 · 2026-04-02 [LATEST]

Input Voltage	Output Voltage	Hold Up Time (ms) 100% Load (10.2 A)	Hold Up Time (ms) 50% Load (5.09 A)
90Vac	51.3	27.5	54.9
	54	27.1	50.9
	57.2	24.7	49.1
264Vac	51.3	27.3	53.3
	54	27.1	51.5
	57.2	25.7	49.1

@ max output voltage @ 50% load, the Hold Up time is a little bit slow. but maybe we don't care....



54V	Remote-Sense (SNS)	OPEN	In Progress

Specification / Procedure

Spec: Up to 500mV total cable-drop compensation, reverse and short circuit protected. Connect SNS+/SNS- leads to point-of-load with twisted pair wire. (Note 1)

Attach sufficiently long wires between the output and the load, which causes a voltage drop exceeding 250mV per wire @ 85% load current. (Rationale for 85%: When remote sense is used, output voltage and output power goes up, when tested with a constant current load. We do not specify operation at P_out>100% rating.)

For instance, the DCR of 1.5m of 12AWG wire is 0.0075ohm. For the 12V model, this yields a drop of 292mV per wire, or 584mV total. Different cable lengths shall be used for other voltage models.

Part A:

- Run at nominal line voltage (e.g. 115Vac)
- Increase the set-point voltage to V_nom + 5% using the potentiometer.

1. Leave the sense leads disconnected from the load. Measure the voltage at the power supply PCB @ 85% load.

2. Connect the twisted-pair sense leads at the point-of-load (at the E-load terminals).

3. Measure the voltage at the power supply PCB again @ 85% load.

The voltage at the PCB should have increased by at least 500mV.

Part B:

1. Detach the SNS leads, reverse the leads (connect SNS+ to -load terminal and SNS- to +load terminal). Check that there is no damage during and after.

Part C:

1. Detach the SNS leads and connect them to each other (short SNS+ to SNS-, alone). Check that there is no damage during and after.

Voltage Model	Gauge	Length (ft)	Resistance	Load (A)	Voltage at Supply	Voltage at Load	Wire Resistance to Meet Drop	Length of Each Wire (ft)	Rounded Length (ft)
	12	1	0.004						
54V	12	8	0.032	8.245	57.2	56.7	0.0606	7.580	8
24V	12	3	0.012	18.552	25.7	25.2	0.0270	3.369	3
12V	12	2	0.008	37.103	13.1	12.6	0.0135	1.684	2

Results

Vision314 · 2026-04-07 [LATEST]
85% load = 8.67A

Part A

Step #	Voltage @ Supply	Voltage @ Load
1	56.83 V	56.12 V
3	57.3 V	56.47 V

Part B - No damage

Voltage @ Supply	Voltage @ Load
58.16 V	57.34V

Part C - No damage

Voltage @ Supply	Voltage @ Load
58V	57.3V

Comments (3)

lauxtim · 2026-04-14
> [RESULT] 85% load = 8.67A
>
> ### Part A
> Step # Voltage @ Supply Voltage @ Load
> 1 56.83 V 56.12 V
> 3 57.3 V 56.47 V

57.3 - 56.83V is 0.47V, which means the remote sense circuit added 470mV to the output voltage, which is still short of 500mV. Question is whether the wire resistance is still too low or if it is indeed limited to ~470mV of added compensation.

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lauxtim · 2026-04-15
See: https://github.com/GlobTek-Engineering/GTM965500P/issues/55#issuecomment-4256097729

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lauxtim · 2026-04-15 [LATEST]
@Vision314

> > [RESULT] 85% load = 8.67A
> > ### Part A
> > Step # Voltage @ Supply Voltage @ Load
> > 1 56.83 V 56.12 V
> > 3 57.3 V 56.47 V
>
> 57.3 - 56.83V is 0.47V, which means the remote sense circuit added 470mV to the output voltage, which is still

short of 500mV. Question is whether the wire resistance is still too low or if it is indeed limited to ~470mV of added compensation.

Actually, we have all the information we need. I apparently just got lost interpreting the data the other day.

Step #	Voltage @ Supply	Voltage @ Load	V_Cable drop
1	56.83V	56.12 V	0.71V
3	57.3 V	56.47 V	
V_Comp	0.47V		
Spec	0.50V (min.)		

The cable/resistor is producing 0.71V of natural drop, but the SNS circuit is only compensating/adding an addition 470mV. Can you double check that Da1 and Da2 (on the regulation board) are the same type as on the 12V and 24V? We didn't happen to swap those out before, right?

It could just be that we are inherently limited at 57.3V due to LLC limitation @ 85% load? Try repeating this test at ~95% load. If the voltage at the supply does not reach 57.3V as it did at 85% load, maybe the LLC has run out of gain and so its inherently limited.

Maybe an interesting point for units running below the resonant frequency (i.e. not a multiple of 12V). They are already operating under boost gain (up the LLC gain curve, to the left of the resonant frequency), so they might not have much boost left.

54V	Output Voltage Fine Adjustment	OPEN	Marked for review

Specification / Procedure

Spec: Via potentiometer, ±5% adjustment range (Note 1)

Run at nominal line voltage (e.g. 115Vac)

Sense leads (SNS) disconnected.

Adjust the potentiometer to minimum and maximum positions. Record the minimum and maximum output voltage, measured at the PCB.

51.3V -> 56.7

Changes Required

Vision314 · 2026-04-06 [LATEST]
Ra8, Ra11 = 200kOhm
Ra9, Ra12 = 4kOhm
Ra10, Ra13 = 10kOhm

Ra14 = 18kOhm

54V	Over-Power Protection (OPP)	CLOSED	Marked for review

Specification / Procedure

Spec: 105-150%, non-latching with auto-recovery

Spec update: Change from "Over-Current Protection" to "Over-Power Protection"

Run at 90Vac and 264Vac.

The specified range applies only for V_out = V_out(nom). (e.g. "12V" @ 12.00V, "24V" @ 24.00V, etc.), voltage measured at the PCB. The nominal design point should be 130 - 135%.

Reference OPP data should also be collected at V_out(-5%) and V_out(+5% + 500mV), for reference although not specified. At V_out(+5% + 500mV), at least 100% output power should be obtainable.

FRANKIE TESTING NOTES:
4/1/2026
Noting current values for regulation board. In order to test Vout max and min (51.3 V -> 57.2V) the following regulation board components were used and give a actual output voltage range of 50.3 V -> 60.0 V

Ra8, Ra11 = 200kOhm
Ra9, Ra12 = 200Ohm
Ra10, Ra13 = 10kOhm

Ra14 = 10kOhm

Changes Required

Vision314 · 2026-04-01 [LATEST]

COMPONENT	OLD VALUE	NEW VALUE
C26	6.8nF	6.8nF
C27	3.3nF	2.2nF

Results

Vision314 · 2026-04-14 [LATEST]

Input Voltage	Output Voltage	OPP Point (A)	OPP Point (%)	PG Time (ms)
90Vac	51.3	15	147%	30
	54	13	128%	10
	57.2	11	114%	10
264Vac	51.3	17	167%	10
	54	13	128%	10
	57.2	11	114%	10

54V	Input Voltage	CLOSED	OK / Resolved

Specification / Procedure

Spec: 90 - 264Vac (Nameplate: 100 - 240Vac)

Confirm that full rated power can be drawn over the entire rated input voltage range. Test full load @ 90Vac, 115Vac, 240Vac, 264Vac

Comments (1)

Vision314 · 2026-04-02 [LATEST]
https://github.com/GlobTek-Engineering/GTM965500P/issues/67#issuecomment-4172429455

Untested — Specifications

54V

54V	Over-Voltage Protection (OVP)	OPEN	In Progress
54V	Short-Corcuit Protection (SCP)	OPEN	In Progress
54V	Component Thermal Stress	OPEN	Unknown
54V	Output Touch Current	OPEN	Unknown
54V	Earth Leakage Current	OPEN	Unknown
54V	Conducted Emissions	OPEN	Unknown
54V	Radiated Emissions	OPEN	Unknown
54V	Electrostatic Discharge (ESD)	OPEN	Unknown
54V	Electrical Fast Transient (EFT)	OPEN	Unknown
54V	Component Electrical Stress	OPEN	Unknown